

Model Context Protocol

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Myfolder: C:\Users\ashok\OneDrive\Documents\crewai\mcp_servers
GitHub link is here

How does MCP help?

The **Model Context Protocol (MCP)** is an open standard and open-source framework introduced by Anthropic in November 2024 to standardize the way artificial intelligence (AI) systems like large language models (LLMs) integrate and share data with external tools, systems, and data sources.[1] MCP provides a standardized interface for reading files, executing functions, and handling contextual prompts

What is MCP?

MCP helps agents to use tools in a standardized way.

Without MCP, when you use a function call directly with an LLM API, you need to:

- ✳ Define model-specific function schemas, which are JSON descriptions of the function, acceptable parameters, and what it returns.
- ✳ Implement handlers (the actual code that executes when a function is called) for those functions.
- ✳ Create different implementations for each model you support.

MCP standardizes this process by:

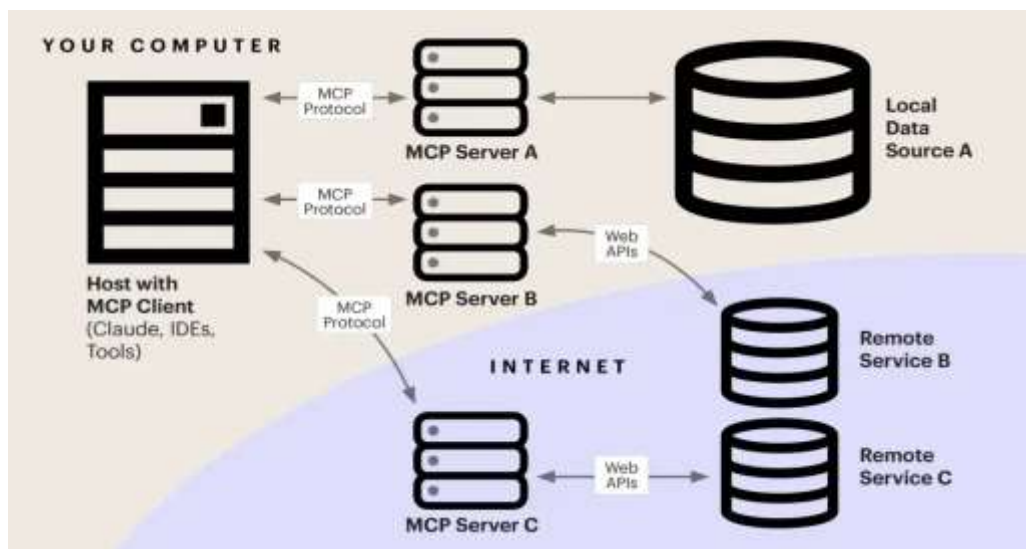
- ✳ Defining a consistent way to specify tools (functions) across any AI system.
- ✳ Providing a protocol for discovering available tools and executing them.
- ✳ Creating a universal, plug-and-play format where any AI app can use any tool without custom integration code.

Key features of MCP are

Key Features

- **Standardized Format:** MCP defines a shared schema for what “context” looks like (e.g., conversations, documents, past interactions).
- **Tool-Agnostic:** It is **not tied to any specific AI model or provider**, making it easier to integrate into any ecosystem (e.g., OpenAI, Anthropic, Hugging Face).
- **Composable:** You can plug in different memory systems, context providers, or agents as long as they follow the protocol.
- **Interoperability:** MCP enables smooth interactions between memory, retrieval-augmented generation (RAG) systems, agents, and LLMs.

How does MCP communicate?



What is MCP Client?

An **MCP Client** is the software component inside an AI application that connects an AI model to external tools and data sources using the **Model Context Protocol (MCP)**.  Google Cloud +1

Core Role

- **The Bridge:** It acts as a translator between an AI assistant (like Claude) and external MCP Servers that hold files, databases, or APIs.
- **The Initiator:** It starts connections, asks servers what tools are available, and sends data requests.
- **The Manager:** It handles the message flow, formatting, and security checks behind the scenes so the AI model does not have to deal with raw code.  YouTube - ByteMonk +3

How It Works

1. **User Prompt:** You type a request into an AI app or developer tool.
2. **Client Action:** The MCP client inside the app recognizes if it needs outside data and queries the correct MCP server.
3. **Response Delivery:** The server replies with the requested data or tool result, which the client formats and hands back to the AI model to write a final answer.  Cloudflare +4

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